Yoson Y

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Objective

Computer Science undergraduate (AI & ML specialization) seeking an AI/IoT robotics internship. Builds ESP32-based robots in C/C++ that integrate IR and motion sensors, motor drivers, servos, and ESP-NOW wireless links, with firmware for sensor calibration, PID control, and motion recording. Also programs in Python and debugs hardware through serial diagnostics.

Education

Christ University, Kengeri Campus — Bangalore, India
B.Tech in Computer Science and Engineering (AI & ML Specialization) — *CGPA: 3.2/4.0* — *Expected Graduation: 2029*

Sacred Heart International School — Marthandam, Tamil Nadu
Class XII, ISC Board — *Percentage: 90.25%* — *2025*

Sacred Heart International School — Marthandam, Tamil Nadu
Class X, ICSE Board — *Percentage: 91%* — *2023*

Technical Skills

- **Languages:** C, C++ (Arduino), Python, Java, SQL
- **Embedded & IoT:** ESP32, Arduino, ESP-NOW, I2C, PWM, Analog Multiplexing, Sensor Calibration
- **Robotics & Electronics:** PID Control, Differential Drive, TB6612FNG, DC & Servo Motors, IR Arrays, MPU6050, RC Receivers, SSD1306 OLED
- **Software & Tools:** Arduino IDE, Git, GitHub, VS Code, Pandas, NumPy, FastAPI, SQLite

Projects

PID Line Follower Robot | Code — *ESP32, C++, TB6612FNG, 8-Channel IR Array*
- Read eight IR sensors through an analog multiplexer with per-sensor min/max calibration, computing line position as a weighted average on a 0–7000 scale.
- Implemented PID steering at a 5 ms sample time, driving two N20 motors through a TB6612FNG with 5 kHz PWM differential speed control.
- Added automatic black/white line detection and line-loss recovery that turns toward the last-seen side.

4-DOF Robotic Arm | Code — *ESP32, C++, Servo Motors, FlySky RC Receiver*
- Decoded four RC receiver channels by pulse-width measurement, rejecting out-of-range pulses and mapping each to joint-specific servo limits.
- Built a teach-and-replay mode that records up to 200 poses and replays them once or in a loop, with stop, slow, and reset controls and LED status.

Gesture-Controlled RC Car | Code — *2x ESP32, MPU6050, ESP-NOW, TB6612FNG*
- Read MPU6050 acceleration over I2C on a transmitter ESP32 and classified hand tilt into five drive commands using a configurable dead zone.
- Sent commands every 70 ms over ESP-NOW to a receiver ESP32 that validates packet size and drives two DC motors through a TB6612FNG.

Certifications

- Introduction to Data Science and AI – IIT Madras, Centre for Outreach and Digital Education (Apr 2025)
- Azure AI Fundamentals – Microsoft, via Certiport (Sep 2026)
- Introduction to Modern AI – Cisco Networking Academy (Sep 2026)
- Programming Fundamentals using Python (Parts 1 & 2), Fundamentals of Responsible AI – Infosys Springboard (May–Jun 2026)

Languages & Interests

- **Hackathons:** Participated in Adobe University Hackathon (Adobe) and InHack Hackathon, Media Meet 2026 (CHRIST University)
- **Languages:** English (Fluent), Tamil (Native), Malayalam